ELEC 231 — Mathematical Methods I for ECE · Fall 2026

# Week 1: General Introduction

Prof. Michael Korenberg · Prepared by Abdelrahman Ramadan
Typed from Prof. Korenberg's Fall 2025 handwritten notes, version 2026-09-13.

This note covers selected content from Chapters 2 and 3 of Murray R. Spiegel's *Applied Differential Equations*, and Chapter 7 and Appendix B of Nilsson & Riedel's *Electric Circuits* (12th ed.).

**Notation.**

| | | | |
|---|---|---|---|
| $\mathrm{j}$ | imaginary unit, $\mathrm{j}^2=-1$ | $\mathcal{L}\{f(t)\}$ | Laplace transform |
| $\mathrm{e}^{-t/\tau}$ | natural exponential | $\mathcal{L}^{-1}\{F(s)\}$ | inverse transform |
| $\frac{\mathrm{d}y}{\mathrm{d}t}$, $\frac{\mathrm{d}^2y}{\mathrm{d}t^2}$ | derivatives | $v * h$ | convolution |
| $u(t)$ | unit step | $\boldsymbol{x}$, $\mathbf{A}$ | vector, matrix |
| $\delta(t)$ | unit impulse | $\tau$ | time constant |

## 1 Complex numbers

Source: Nilsson & Riedel, *Electric Circuits*, Appendix B (the professor's own sub-heading on this page).

### 1.1 Rectangular form

$$z = a + \mathrm{j}b.$$

Here $a$ is the *real part*, $b$ is the *imaginary part*, and $\mathrm{j} = \sqrt{-1}$. Both $a$ and $b$ are real, and

$$a = \mathrm{Re}\{z\}, \qquad b = \mathrm{Im}\{z\}.$$

### 1.2 Polar form

$$z = c\,\mathrm{e}^{\mathrm{j}\theta},$$

where $c$ is the magnitude, $c = |z|$; e is the base of the natural logarithm; and $\theta$ is the angle, $\theta = \angle z$. In angle notation,

$$z = c\angle\theta.$$

### 1.3 Euler's identity

$$\mathrm{e}^{\mathrm{j}\theta} = \cos\theta + \mathrm{j}\sin\theta, \tag{1.1}$$

so that

$$c\,\mathrm{e}^{\mathrm{j}\theta} = c\cos\theta + \mathrm{j}c\sin\theta^{\mathrm{C1}} = a + \mathrm{j}b.$$

Euler's identity makes the polar form equal to the rectangular form. Indeed, to convert the polar

[C1] Corrected from the handwritten original, which reads: "$a\cos\theta + \mathrm{j}c\sin\theta$" — the next line of the same display sets this equal to $a + \mathrm{j}b$ with $a = c\cos\theta$, so the first term is $c\cos\theta$; the $c$ is written like an $a$. Source: Complex_numbers_and_ODEs:1.

form $z = c\,\mathrm{e}^{\mathrm{j}\theta}$ to the rectangular form $z = a + \mathrm{j}b$, set

$$a = c\cos\theta, \qquad b = c\sin\theta.$$

## 2 Trigonometric identities that follow from Euler's identity

Starting from $\mathrm{e}^{\mathrm{j}A}\,\mathrm{e}^{\mathrm{j}B} = \mathrm{e}^{\mathrm{j}(A+B)}$ and writing each exponential with Eq. (1.1),

$$(\cos A + \mathrm{j}\sin A)(\cos B + \mathrm{j}\sin B) = \cos(A+B) + \mathrm{j}\sin(A+B)$$
$$\cos A\cos B - \sin A\sin B + \mathrm{j}\big(\sin A\cos B + \cos A\sin B\big) = \cos(A+B) + \mathrm{j}\sin(A+B).$$

Equating real parts of the left and right sides:

$$\cos(A+B) = \cos A\cos B - \sin A\sin B.$$

Equating imaginary parts of the left and right sides:

$$\sin(A+B) = \sin A\cos B + \cos A\sin B.$$

### 2.1 Approximate outline only

Consider the equation

$$\frac{\mathrm{d}y(t)}{\mathrm{d}t} + y(t) = 0, \qquad y(0) = y_0.$$

Replacing the derivative by a backward difference,

$$\frac{y(t) - y(t-\Delta)}{\Delta} + y(t) \cong 0 \qquad \text{for small } \Delta,$$

so that

$$y(t)\left(1 + \frac{1}{\Delta}\right) = \frac{y(t-\Delta)}{\Delta}, \qquad \text{i.e.} \qquad y(t)(1+\Delta) = y(t-\Delta).$$

Let $k = t/\Delta$. Then

$$y(t)\,(1+\Delta)^{t/\Delta} = y(0), \qquad \mathrm{e} = \lim_{\Delta\to 0}(1+\Delta)^{1/\Delta},$$

and therefore

$$y(t) = y(0)\,\mathrm{e}^{-t}.$$

**Added explanation (not in the 2025 notes).** The professor posted a longer version of this outline on a separate sheet; it is transcribed unchanged as Section 11 at the end of these notes. Only this cross-reference has been added.

## 3 Converting rectangular form to polar form

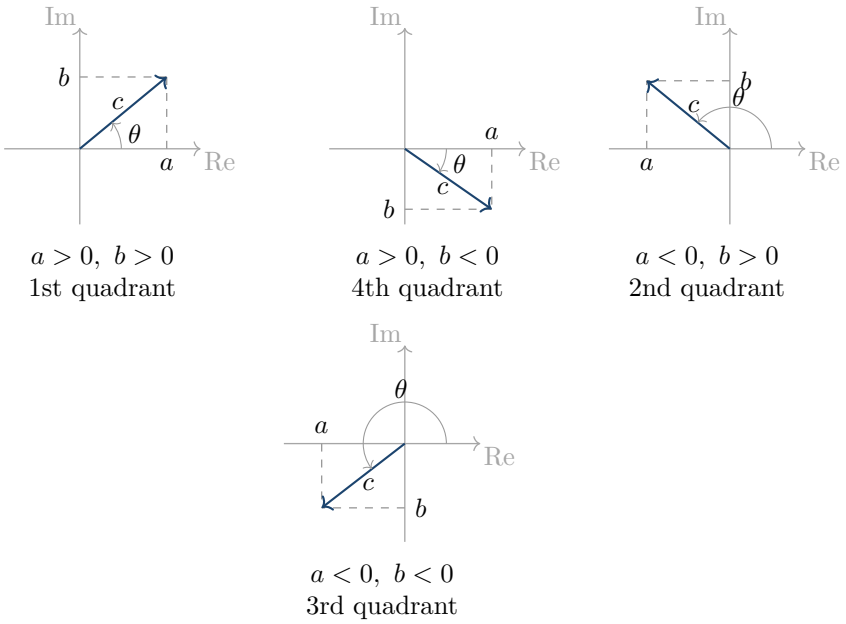



**Figure 1.1.** The four sketches of page ③: the angle $\theta$ of $z = a + \mathrm{j}b$ depends on the quadrant in which $z$ lies.

$$c = \sqrt{a^2 + b^2} = |z|.$$

To convert the rectangular form $z = a + \mathrm{j}b$ to the polar form $z = c\,\mathrm{e}^{\mathrm{j}\theta}$, set

$$c = \sqrt{a^2 + b^2}, \qquad \tan\theta = \frac{b}{a},$$

where $\theta$ depends on the quadrant where the complex number $z$ is located (Figure 1.1).

## 4 Arithmetic operations

### 4.1 Addition or subtraction

Put the complex numbers in rectangular form:

$$\begin{aligned} z_1 + z_2: &\quad a_1 + \mathrm{j}b_1 + a_2 + \mathrm{j}b_2 = (a_1 + a_2) + \mathrm{j}(b_1 + b_2), \\ z_1 - z_2: &\quad a_1 + \mathrm{j}b_1 - (a_2 + \mathrm{j}b_2) = (a_1 - a_2) + \mathrm{j}(b_1 - b_2). \end{aligned}$$

### 4.2 Multiplication and division

Put the complex numbers in polar form:

$$\begin{aligned} z_1 \cdot z_2: &\quad c_1\mathrm{e}^{\mathrm{j}\theta_1} \cdot c_2\mathrm{e}^{\mathrm{j}\theta_2} = c_1c_2\,\mathrm{e}^{\mathrm{j}(\theta_1+\theta_2)}, \\ &\frac{z_1}{z_2} = \frac{c_1}{c_2}\,\mathrm{e}^{\mathrm{j}(\theta_1-\theta_2)}, \qquad z_2 \neq 0. \end{aligned}$$

### 4.3 Complex conjugate

The complex conjugate of $z$ is denoted by $z^*$; for example

$$(a+\mathrm{j}b)^* = a-\mathrm{j}b, \qquad \left(c\,\mathrm{e}^{\mathrm{j}\theta}\right)^* = c\,\mathrm{e}^{-\mathrm{j}\theta}.$$

Note that

$$\begin{aligned} z\cdot z^* &= a^2+b^2 = c^2,\\ z+z^* &= 2a = 2\,\mathrm{Re}\{z\},\\ z-z^* &= (a+\mathrm{j}b)-(a-\mathrm{j}b) = 2\mathrm{j}b = \mathrm{j}\,2\,\mathrm{Im}\{z\}. \end{aligned}$$

### 4.4 Multiplication of numbers in rectangular form

$$(a_1+\mathrm{j}b_1)(a_2+\mathrm{j}b_2) = a_1a_2 - b_1b_2 + \mathrm{j}\,(a_1b_2+a_2b_1).$$

### 4.5 Division of numbers in rectangular form

**Example 1.1 (Division in rectangular form).** Multiply numerator and denominator by the conjugate of the denominator:

$$\begin{aligned} \frac{80+\mathrm{j}18}{8+\mathrm{j}10} &= \frac{(80+\mathrm{j}18)(8-\mathrm{j}10)}{(8+\mathrm{j}10)(8-\mathrm{j}10)} \qquad z_2z_2^* = |z_2|^2 \text{ is real}\\ &= \frac{640+180+\mathrm{j}(144-800)}{64+100}\\ &= \frac{820-\mathrm{j}656}{164}\\ &= 5-\mathrm{j}4. \end{aligned}$$

## 5 Integer power of a complex number (de Moivre's theorem)

$$z^k = (a+\mathrm{j}b)^k = \left(c\,\mathrm{e}^{\mathrm{j}\theta}\right)^k = c^k\,\mathrm{e}^{\mathrm{j}k\theta} = c^k\cos(k\theta)+\mathrm{j}\,c^k\sin(k\theta), \qquad k=\text{integer}.$$

## 6 Absolute value (magnitude) of $\mathrm{e}^{\mathrm{j}\theta}$

$$|\mathrm{e}^{\mathrm{j}\theta}| = |\cos\theta+\mathrm{j}\sin\theta| = \sqrt{\cos^2\theta+\sin^2\theta} = 1.$$

## 7 Roots of a complex number

$$(z)^{1/k} = \left(c\,\mathrm{e}^{\mathrm{j}\theta}\right)^{1/k} = c^{1/k}\times \mathrm{e}^{\mathrm{j}\theta/k}\times x_\ell, \qquad k=\text{integer},$$

where $x_\ell$, $\ell = 0,1,\ldots,k-1$[C2], are the $k$-th roots of 1. Each root $x_\ell$ has magnitude 1, and angle

$$\theta_\ell = \frac{2\pi}{k}\,\ell \text{ radians} = \frac{360^\circ\,\ell}{k} \text{ degrees}, \qquad \ell = 0,\ldots,k-1.$$

[C2] Corrected from the handwritten original, which reads: "$\ell = 1,\ldots,k$" — the next clause of the same sentence gives $\ell = 0,\ldots,k-1$, and the worked example below uses $\ell = 0,1,2,3$ for $k=4$. Source: Complex_numbers_and_ODEs:6.

**Example 1.2 (The four fourth roots of a complex number).** Let $z = 81\,\mathrm{e}^{\mathrm{j}60^\circ}$ and $k = 4$, so that $c = 81$ and $\theta = 60^\circ$. The fourth roots of 1 are

$$x_\ell = \mathrm{e}^{\mathrm{j}\,360^\circ \ell/k} = \mathrm{e}^{\mathrm{j}\,90^\circ \ell}, \qquad \ell = 0, \ldots, 3.$$

Hence

$$\begin{aligned}(z)^{1/4} &= 81^{1/4}\,\mathrm{e}^{\mathrm{j}\,60^\circ/4}\,x_\ell, \qquad \ell = 0, \ldots, 3 \\ &= 3\,\mathrm{e}^{\mathrm{j}\,15^\circ}\,\mathrm{e}^{\mathrm{j}\,90^\circ \ell}, \qquad \ell = 0, \ldots, 3,\end{aligned}$$

so the fourth roots of $81\,\mathrm{e}^{\mathrm{j}60^\circ}$ are

$$3\,\mathrm{e}^{\mathrm{j}15^\circ}, \quad 3\,\mathrm{e}^{\mathrm{j}105^\circ}, \quad 3\,\mathrm{e}^{\mathrm{j}195^\circ}, \quad 3\,\mathrm{e}^{\mathrm{j}285^\circ}.$$

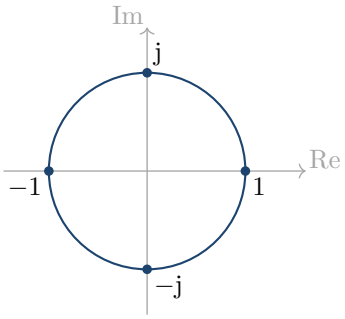



**Figure 1.2.** The fourth roots of 1: $x_\ell = \mathrm{e}^{\mathrm{j}90^\circ \ell}$, $\ell = 0, 1, 2, 3$, i.e. 1, j, −1, −j.

## 8 Natural response of first-order RL and RC circuits

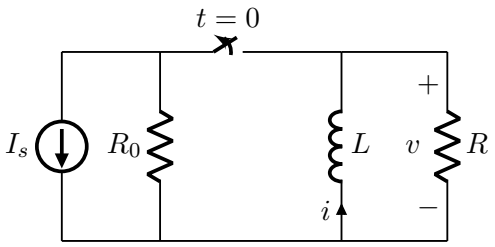



**Figure 1.3.** Fig. 1: the switch has been closed for a long time and opens at $t = 0$. Find $v(t)$ and $i(t)$, $t \geq 0$.

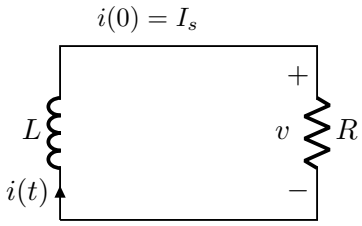



**Figure 1.4.** Fig. 2: the circuit for $t \geq 0$, with $i(0) = I_s$.

The switch has been closed for a long time and then opens at $t = 0$. Find $v(t)$ and $i(t)$, $t \geq 0$. For $t \geq 0$ only the loop of Figure 1.4 is left, and Kirchhoff's voltage law around it gives a first-order differential equation with constant coefficients:

$$\begin{aligned} R\,i + L\,\frac{\mathrm{d}i}{\mathrm{d}t} &= 0, \\ \frac{\mathrm{d}i}{\mathrm{d}t} + \frac{R}{L}\,i &= 0, \\ \frac{\mathrm{d}i}{i} &= -\frac{R}{L}\,\mathrm{d}t, \\ \int_{i(t_0)}^{i(t)} \frac{\mathrm{d}x}{x} &= -\frac{R}{L}\int_{t_0}^{t} \mathrm{d}y. \end{aligned}$$

Here $t_0 = 0$, so

$$\begin{aligned} \ln\big(i(t)\big) - \ln\big(i(0)\big) &= -\frac{R}{L}\,t, \\ \ln\frac{i(t)}{i(0)} &= -\frac{R}{L}\,t, \\ \frac{i(t)}{i(0)} &= \mathrm{e}^{-(R/L)t}, \\ i(t) &= i(0)\,\mathrm{e}^{-\frac{R}{L}t}. \end{aligned}$$

**Note.** Two facts are written in the margin of this page and are used throughout the course: *the inductor current cannot change instantaneously*, $i(0^-) = i(0^+)$; and *in dc steady state the inductor acts like a short.*

## 9 Tutorial problem

**Problem 1 (Tutorial problem, Week 1).** If $L = 8\,\mathrm{mH}$ and $R = 4\,\Omega$ in Fig. 2, and $I_s = 4\,\mathrm{A}$, what is the numerical expression for $i(t)$, $t \geq 0$? Sketch $i(t)$, $t \geq 0$.

**Remark.** This is the problem set as Pre-tutorial 1 of the course. The worked answer — $\tau = L/R = 2\,\mathrm{ms}$ and $i(t) = 4\mathrm{e}^{-500t}$ A, $t \geq 0$ — is the professor's own, written on page ① of the Week 2 sheets; it is typed in the Pre-tutorial 1 documents of this package and is not repeated here.

## 10 Time constant

$$\tau = \frac{L}{R},$$

$$i(t) = I_s\,\mathrm{e}^{-t/\tau}, \qquad t \geq 0, \tag{1.2}$$

where $I_s$ is the value of the initial current in the inductor, after the switch has opened.

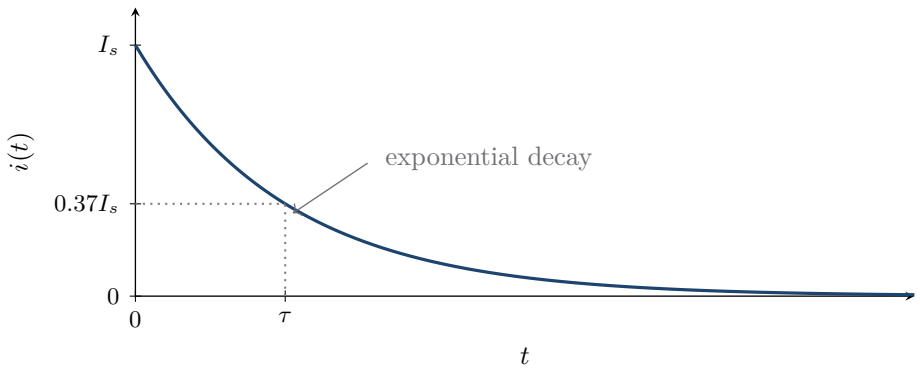



**Figure 1.5.** The sketch of page ⑧: $i(t) = I_s\mathrm{e}^{-t/\tau}$ decays from $I_s$ at $t = 0$.

One time constant after the switch has opened, the current in the inductor is $I_s\mathrm{e}^{-1} \approx 0.37\,I_s$.

## 11 Expanded derivation (not rigorous)

$$\frac{\mathrm{d}y}{\mathrm{d}t} + y(t) = 0, \qquad y(0) = y_0.$$

For small $\Delta$,

$$\frac{y(t) - y(t-\Delta)}{\Delta} + y(t) \approx 0, \qquad \text{so} \qquad y(t)\left(1 + \frac{1}{\Delta}\right) \approx \frac{y(t-\Delta)}{\Delta}.$$

Consider

$$y(t)\,(\Delta + 1) = y(t - \Delta). \tag{1.3}$$

Then, stepping backwards through Eq. (1.3),

$$\begin{aligned}
t = \Delta \text{ in Eq. (1.3)} &\implies y(0) = y(\Delta)(1+\Delta), \\
t = 2\Delta \text{ in Eq. (1.3)} &\implies y(\Delta) = y(2\Delta)(\Delta+1) \implies y(0) = y(2\Delta)(1+\Delta)^2, \\
t = 3\Delta \text{ in Eq. (1.3)} &\implies y(3\Delta)(\Delta+1) = y(2\Delta), \\
y(0) &= y(2\Delta)(1+\Delta)^2 = y(3\Delta)(\Delta+1)^3, \quad \text{etc.}
\end{aligned}$$

Let $k = t/\Delta$. Then

$$\begin{aligned}
y(0) &= y(k\Delta)(\Delta+1)^k = y(t)\,(\Delta+1)^k, \\
y(t) &= y(0)\,(\Delta+1)^{-k} = y(0)\,(\Delta+1)^{\frac{1}{\Delta}(-t)}.
\end{aligned}$$

Since

$$\mathrm{e} = \lim_{\Delta \to 0} (1+\Delta)^{1/\Delta},$$

this gives

$$y(t) = y(0)\,\mathrm{e}^{-t}, \qquad t \geq 0.$$